Sec: SR.IIT_*COSC(MODEL-A&APEX) | GTM-8 | Date: 17-03-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**C**	2	**C**	3	**D**	4	**C**	5	**C**
6	**A**	7	**B**	8	**B**	9	**B**	10	**B**
11	**A**	12	**D**	13	**D**	14	**C**	15	**C**
16	**A**	17	**B**	18	**D**	19	**C**	20	**C**
21	**5**	22	**6**	23	**3**	24	**30**	25	**125**

CHEMISTRY

26	**D**	27	**C**	28	**D**	29	**D**	30	**D**
31	**A**	32	**B**	33	**B**	34	**B**	35	**B**
36	**A**	37	**A**	38	**A**	39	**C**	40	**D**
41	**A**	42	**B**	43	**D**	44	**D**	45	**B**
46	**2**	47	**33**	48	**87**	49	**6**	50	**454**

MATHEMATICS

51	**D**	52	**B**	53	**A**	54	**D**	55	**C**
56	**A**	57	**B**	58	**B**	59	**B**	60	**B**
61	**C**	62	**C**	63	**B**	64	**A**	65	**C**
66	**A**	67	**A**	68	**B**	69	**B**	70	**A**
71	**5**	72	**28**	73	**3**	74	**2400**	75	**4**

For More Join: @IITJEE_Advanced

SOLUTIONS

PHYSICS

1. $\frac{(8\times10^3)^2}{R_p}=80\times10^3$

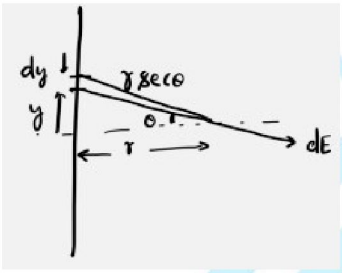
$R_p=800\Omega$

$\frac{(160)^2}{R_s}=80\times10^3 \Rightarrow R_S=0.32\Omega$

2. $v_T=\frac{2}{9}R^2\frac{g}{n}\left(\rho_{sph}-\rho_{Liquid}\right)$

3. Conceptual

4.

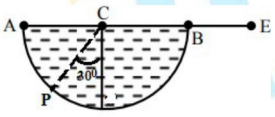


$dq=\lambda\,dy \quad \tan\theta=\frac{y}{r}\Rightarrow y=r\tan\theta \quad dy=r\sec^2\theta\,d\theta$

$dE_{\parallel}=dE\cos\theta=\frac{k\lambda_0}{r}\theta\cos\theta d\theta$

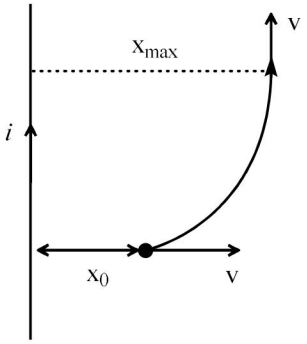
$E_{\parallel}=\frac{k\lambda_0}{r}\int_{-\beta}^{\alpha}\theta\cos\theta d\theta=\frac{k\lambda_0}{r}[\alpha\sin\alpha+\cos\alpha-\beta\sin\beta-\cos\beta)]$

5.



$\mu\sin60^0=1\times\sin90^0 \quad \mu=\frac{2}{\sqrt{3}}$

6. Let the particle be initially at P. Take the wire as the y-axis and the foot of perpendicular from P to the wire as the origin. Take the line OP as the x-axis. We have, OP=xo. The magnetic field B at any point to the right of the wire is along the negative z-axis. The magnetic force on the particle is, therefore, in the x-y plane. As there is no initial velocity along the z-axis, the motion will be in the x-y plane. Also, its speed remains unchanged. As the magnetic field is not uniform, the particle does not go along a circle. The force at time t is $F=q\left(\vec{v}\times\vec{B}\right)$



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$$\vec{F} = q\left(v_x\hat{i} + v_y\hat{j} + v_z\hat{k}\right) \times \left(-\frac{\mu_0 i}{2\pi x}\hat{k}\right)$$

Thus, $a_y = \frac{dv_y}{dt} = \frac{q\mu_0 i}{2\pi m x}\frac{dy}{dt} = \int_0^v dv_y = \frac{q\mu_0 i}{2\pi m}\int_{x_0}^{x_{max}} \frac{dx}{x}$

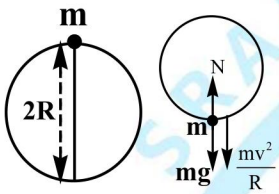
$$x_{max} = x_0 e^{\frac{2\pi m v}{q\mu_0 i}}$$

7. $T = \frac{T_1 + T_2}{2} = \frac{227 + 127}{2} = 177^0 c$

8. Logic: Both statements are scientifically true. However, R explains why sound should be slower in steel. The real reason A is true is that the Bulk Modulus of steel increases much more drastically than its density does, overriding the density factor.

9. Conceptual

10.



Work energy theorem

$$mg(2R) + \frac{1}{2}kR^2 = \frac{1}{2}mv^2$$

$$v = \sqrt{4gR + \frac{kR^2}{m}}$$

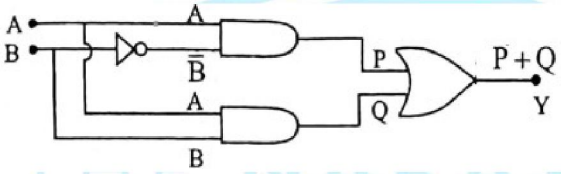
at bottom point, string will become slack

$$N = mg + \frac{mv^2}{R} = 5mg + kR$$

11. Conceptual

12. Option (D) is the correct reaction.

13



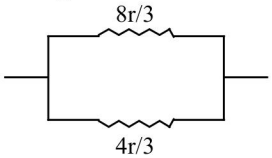
$P = A.\bar{B}$

$Q = A.B$

$Y = P + Q = A.\bar{B} + A.B = A.(B + \bar{B}) = A.I$

$Y = A$

14. Using balanced Wheatstone's network we can simplify the circuit as below.



$$R_{eq} = \frac{8r}{9}$$

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15. Charge flowing through both capacitor is same

$\because U = \frac{q^2}{2C}$

$\because q = CV \Rightarrow CV = \text{constant}$

From the given graph $C_1 > C_2$ and Hence $U_2 > U_1$

16. Energy density $= \frac{1}{2}\epsilon_0 E^2$

(Energy)$_1$ = (Energy)$_2$ (Given)

$$\left(\frac{9}{16}\right)\frac{1}{2}\epsilon_0 E_1^2 \pi R_1^2 \times L_1 = \frac{1}{2}\epsilon_0 E_2^2 \frac{4}{3}\pi R_2^3$$

17. $u_1 + k_1 = u_2 + k_2$

$$-PE\cos 60 + 0 = -PE\cos 0 + \frac{1}{2}I\omega^2$$

$$\frac{1}{2}I\omega^2 = \frac{PE}{2} \ (\text{or}) \ \omega = \sqrt{\frac{PE}{I}}$$

18. $\lambda_P = \lambda_{ph} = \lambda = \frac{hc}{E} \quad E = hc/\lambda$

$$KE = \frac{1}{2}m_p v^2 = \frac{p^2}{2m_p} = \frac{(h/\lambda)^2}{2m_p} = \frac{1}{2m_p}\left(\frac{E}{C}\right)^2$$

$$KE = \frac{E^2}{2m_p c^2}$$

19. $x_{am} = \frac{\int x dm}{\int dm} = \frac{\int x\lambda dx}{\int \lambda dx} = \frac{\int \lambda_0 x^3 dx}{\int \lambda_0 x^2 dx}$

$$x_{am} = \frac{x^4/4}{x^3/3} = \frac{3}{4}x = \frac{3}{4}\ell$$

20. $\frac{1}{1} = \frac{100}{f} + \frac{100}{20}$

$f = 5f + 100$

$4f = -100$

$f = -25$

21. $V = \pi r^2 l$

$\frac{dV}{V} = 2\frac{dr}{r} + \frac{dl}{l}$

$LC = \frac{Pitch}{no.of \ divisions} = \frac{0.75}{15} mm$

$0.05 \ mm$

Thus, $\frac{dV}{V} = 2\left(\frac{0.05}{2.5}\right) + \left(\frac{0.05}{5}\right) = \frac{4}{100} + \frac{1}{100} = \frac{5}{100}$

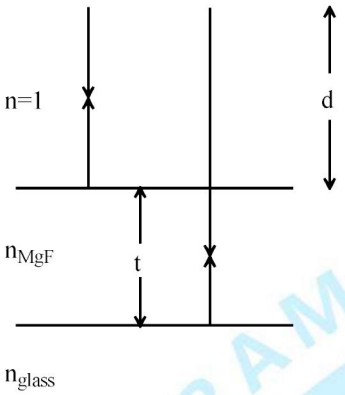
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22. $\frac{35}{22}\left(\frac{3}{2}MR^2+\frac{5}{4}MR^2\right)=\frac{7}{5}MK^2R^2$

$\Rightarrow K^2=\frac{25}{8}$

$\sqrt{8K^2}+1=6$

23.



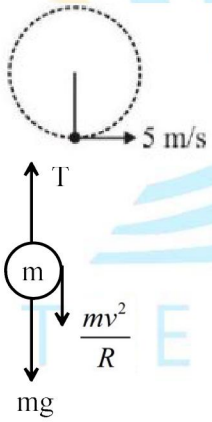
$X_1=2d+\frac{\lambda}{2}$

$X_2=2d+2\mu t+\frac{\lambda}{2}$

$X_2-X_1=2\mu t=\frac{\lambda}{2}$

$\mu=\frac{\lambda}{4t}=\frac{500}{400}=1.25$

24.



$T=mg+\frac{mv^2R}{}$

$\frac{T}{YA}=strain$

$\text{strain}=\frac{mg+\frac{mv^2}{R}}{AY}=20+\frac{\frac{2(5)^2}{0.5}}{4\times10^{-6}\times10^{11}}=30\times10^{-5}$

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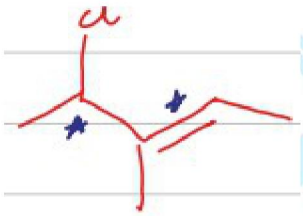
25. $F = xA\frac{dv}{dy}$

$$\frac{x}{F} = \frac{1}{A\frac{dv}{dy}} = \frac{[L]}{[L^2][LT^{-1}]} = [M^0L^{-2}T^1]$$

CHEMISTRY

27. $C-O \quad 143\,pm \quad (A)$
$N-O \quad 136\,pm \quad (C)$
$N=O \quad 122\,pm \quad (D)$
$C=O \quad 121\,pm \quad (B)$

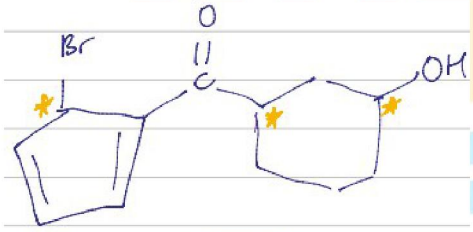
28.



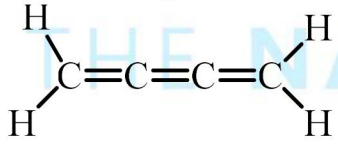
29. NCERT text

30. $\lambda_m = \wedge^0 m - A\sqrt{C}$
for strong electrolyte $\rightarrow$ straight line
for weak electrolyte $\rightarrow$ Hyperbolic curve
$W \alpha Q$ {According to faraday's Law}
$W = ZQ$

31.



A) Alcohols are insoluble in bicarbonates
B) This compound contains 3 chiral carbon. Hence it is optically active



Odd cumulene are planes

32. Statement-I: Dipole moment order
$C-Cl > C-F > C-Br > C-I$
Statement-II: PbI_2 exists but PbI_4 does not exist due to low electronegativity of Iodine

33. $PbSO_4 + CH_3COONH_4 \rightarrow (NH_4)_2[Pb(CH_3COO)_4] + (NH_4)_2SO_4$

$(NH_4)_2[Pb(CH_3COO)_4] \Rightarrow$ Ammonium tetraacetoplumbate (II)

34.

Series	n_1	n_2
(A) Paschen	3	4
(B) Balmer	2	5

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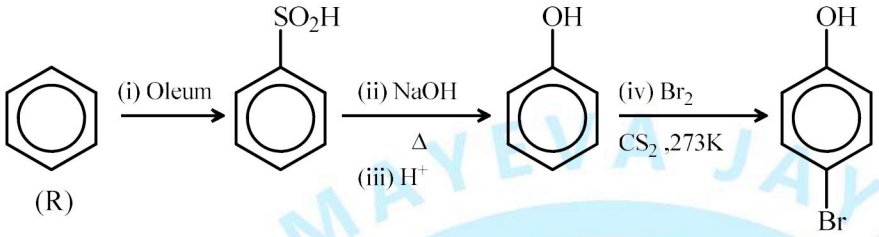
(C) Paschen 3 5
(D) Bracket 4 Q

$$\Delta E = 13.6Z^2\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$$

$\Rightarrow$ Correct order $A < C < B < D$ (Energy)

35. Ans: A-II,B-I,C-III,D-IV
Conceptual

36.



37.

$$MnO_2 + 4HCl \longrightarrow Mncl_2 + Cl_2 + 2H_2O$$

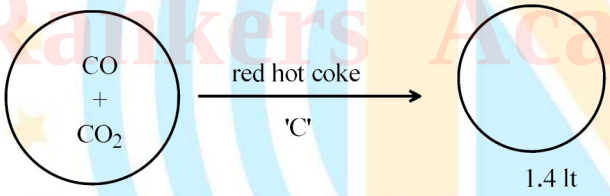
$\frac{17.4}{87}$ *excess*

0.2 mole 0.2 mole Cl_2

Weight of Cl_2 obtained = 0.2×71

$= 14.2g$

38.



$\Rightarrow$

- $CO + C \longrightarrow no\ reaction$
- $CO + C \longrightarrow 2(CO)$

$1:1:2\left(same\ vol^m\ ratio\right)$.

Let $'x'\ell t.$ of CO is taken

$(1-x)\ell f.$ of CO_2

$\rightarrow$ CO will remain unaffected

$CO_2 + C \longrightarrow 2CO$

$(1-x)\ell t$ - -

0 $2(1-x)\ell t.$

$x + 2(1-x) = 1.4$

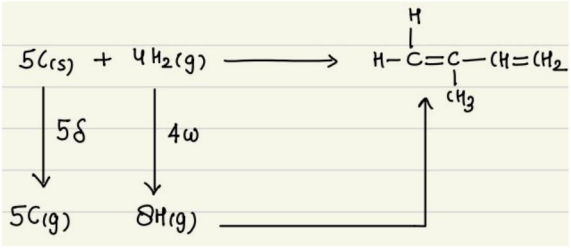
$x + 2 - 2x = 1.4$

$x = 0.6\ell t \longrightarrow$ Volume of $CO_{(g)}$

So, $CO_2 = 1 - x = 0.4\ell t.$

For More Join: @IITJEE_Advanced

39.



$\Delta H_F^{\bullet} = (5\delta + 4w) - (8\alpha + 2\beta + 2\gamma)$

40. Solution

Based on the above analysis

- A (Reimer-Tiemann reaction) $\rightarrow$ III $(CHCl_3 + aq.NaOH)$

- B (Birch reduction) $\rightarrow$ $I\,(Na, liq.NH_3 + EtOH)$

-C (Rosenmund reaction) $\rightarrow$ $IV\,(H_2, pd - BaSO_4)$

-D (Friedel-Crafts reaction) $\rightarrow$ $II\,(CH_3Cl + anhydrous\ AlCl_3)$

Final Answer

The correct option is $(1)\ A - III, B - I, C - IV, D - II.$

41. (a) $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

$t = 0 \quad 3.0M$

$t = 30 \quad 2.75$

$$\frac{-\Delta[N_2O_5]}{\Delta t} = \frac{0.25}{30}$$

$$\frac{1}{2} \times \frac{-\Delta[N_2O_5]}{\Delta t} = \frac{1}{4} \times \frac{\Delta[NO_2]}{\Delta t}$$

$$\frac{\Delta[NO_2]}{\Delta t} = \frac{0.25}{30} \times 2 = 1.66 \times 10^{-2} M \text{ / min}$$

42. (d) A-II,B-III,C-I,D-IV

Starch: $\alpha\, glucose$

Cellulose: $\beta\, glucose$

Nucleic acid: Nucleotide

Protein: α amino acids

43. Ans: d $\left(CH_3C\ddot{O}O > Ph\ddot{O} > O\ddot{H} > R\ddot{O}\right)$

44. Rate of $SN' \propto$ stability of carbocation

46. $\underset{0.1-x}{H_2} \times \rightleftharpoons \underset{x+y}{H^+} + \underset{x-y}{HX^-} \qquad k_{a_1} = \dfrac{[HX^-][H^+]}{[H_2X]}$

$$1.5 \times 10^{-10} = \frac{(x-y)(x+y)}{0.1-x}$$

$$\underset{x-y}{H} \times \rightleftharpoons \underset{y+x}{H^+} + \underset{y}{X^{-2}} \qquad k_{a_2} = \frac{[H^+][X^{-1}]}{[HX^-]}$$

$$1.8 \times 10^{-16} = \frac{(x+y)(y)}{(x-y)}$$

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Since $k_{a_1} >> K_{a_2}$

$x - y\,ax \,\&\, x + y \approx x$

$1.8\times10^{-16} = \dfrac{x.y}{x}$

$y = 1.8\times10^{-16}$

$\left[X^{-2}\right] = y = 1.8\times10^{-16}$

47. $a = 12, b = 9, c = 12$

48. no.of moles of NaCl per line =0.1

$\pi = 4.6 atm$

$T = 300 K$

If Nacl is not dissociated

$\pi_{normal} = CRT = 0.1\times0.082\times300$

$= 2.46 atm$

$\pi_{observed} = 4.6 atm$

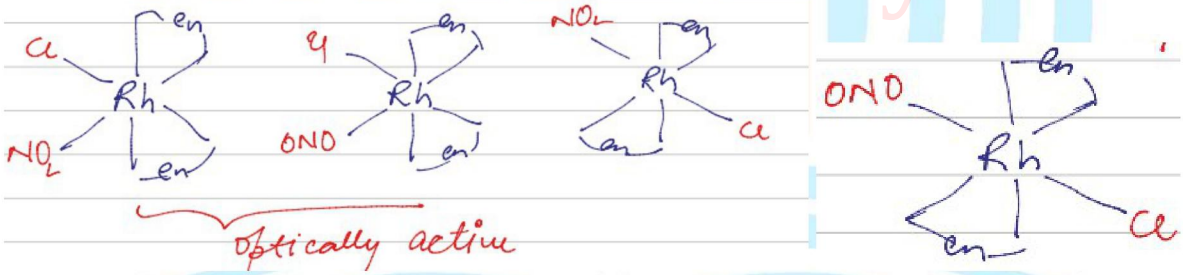
$i = \dfrac{observed\ colligative\ property}{Normal\ colligative\ property}$

$i = \dfrac{4.6}{2.46} = 1.87$

$\alpha = \dfrac{i-1}{n-1} \Rightarrow \dfrac{1.87-1}{2-1} = 0.87$

$\%\alpha = 87\%$

49.



50. $E_{cell} = E^0_{cell} + \dfrac{2.303\,RT}{n_f}\log k_{sp}$

$= 0.49 + \dfrac{0.059}{1}\log10^{-16}$

$= 0.49 - 16\times0.059 = 0.454 = 454 mV$

MATHS

51. $2\tan^{-1}\left(\sin^3 x\right) = \tan^{-1}\left(\dfrac{2\sin^3 x}{1-\sin^6 x}\right) = \tan^{-1}\left(\tan^2 x\right)$

$\Rightarrow \dfrac{2s^3}{1-s^6} = \dfrac{s^2}{1-s^2}$

$\Rightarrow \sin x = 0$ (or)

$2s = 1 + s^2 + s^4$

$\Rightarrow s^4 + (s-1)^2 = 0$ (Not Possible)

For More Join: @IITJEE_Advanced

52. For $a=b=c=1 \Rightarrow y=x^2+x+1$ passes through $(0,1)$.

$(0,0) \in R_3$ as $0<c$ and $(2,1) \in R_3$ as $2<a+b+c$

53. Conceptual

54. $1 \le f(1)<f(3)<f(5) \le 7 \rightarrow {}^7C_3 = 35$

$f(2)<f(4) \rightarrow {}^4C_2 = 6$

$\Rightarrow 35 \times 6 = 210$

55. Minimem distance $= 2-\sqrt{2}$

Maximum distance $= 2$

by plotting graph.

Sum $= 4-\sqrt{2}$

56. The image of $A(1,2)$ across the x-axis is $A'(1,-2)$. By the laws of reflection, the reflected ray travels along the line connecting the virtual image A' and the target $b(5,3)$. The equation of line $A'B$ is

$: y-3=\frac{3-(-2)}{5-1}(x-5)$

The point of reflection P is the x-intercept. Setting y=0 yields $x=\frac{13}{5}$. Therefore, P is $x=\left(\frac{13}{5},0\right)$.

57. If $x>0 \Rightarrow \ln(1+x)-\frac{x+1}{1+x}+\frac{1}{1+x}$.

$f'(x)=\frac{1}{1+x}-\frac{1}{(1+x)^2}=\frac{x}{(1+x)^2}$.

If $x<0 \rightarrow \ln(1-x)+\frac{x+1}{1+x}-\frac{1}{1+x}=\frac{-x^2-3x}{(1-x)(x+1)^2}$

$\Rightarrow$ Stationary Points: $x=0,-3$

Extremum at $x=-3$

58. The domain $x^2 \le y \le x$ restricts the region to $x,y \in [0,1]$. The condition $[x+y]=0$ implies $0 \le x+Y \le 1$, So $y<1-x$. The bounded region lies below $\min(x,1-x)$ and above $y=x^2$.The intersection of y=x and y=1-x is $x=1/2$. The intersection of $y=x^2$ and $y=1-x$ Is $x=\frac{\sqrt{5}-1}{2}=\alpha$

$\text{area}=\int_0^{1/2}(x-x^2)dx+\int_{12}^{\alpha}(1-x-x^2)dx$

$\text{Area}=\left[\frac{x^2}{2}-\frac{x^3}{3}\right]_0^{1/2}+\left[x-\frac{x^2}{2}-\frac{x^3}{3}\right]_{1/2}^{\alpha}=\frac{5\sqrt{5}-10}{12}$

59. If all are odd $\rightarrow (3,3,1),(1,9,1) \rightarrow 6$ cases

If 2 are even & lis odd

$\rightarrow 2x+2y+(2z-1)=9$

$\Rightarrow x+y+z=5 \rightarrow 6$ ways

$6 \times 3 = 18$ ways

$\text{Probability}=\frac{6}{6+18}=\frac{1}{4}$

60. It can touch $x=0$, $y=x$ or it can pass through origin

For More Join: @IITJEE_Advanced

61. For $y^2=8x, a=2$. Let the extremities of the focal chord be $P\left(2t^2,4t\right)$ and $Q\left(2/t^2,-4t\right)$. The area of ΔOPQ with vertex $O(0,0)$ is :

$$\text{Area} = \frac{1}{2}\left|\left(2t^2\right)\left(-\frac{4}{t}\right)-\left(\frac{2}{t^2}\right)(4t)\right| = 4\left|t+\frac{1}{t}\right|$$

Equating to $12\sqrt{2}$ yields $\left|t+\frac{1}{t}\right|=3\sqrt{2}$ The length of chord is given by $L=a\left(t+\frac{1}{t}\right)^2$

$$L=2\left(3\sqrt{2}\right)^2=2(18)=36$$

62. $$a^2-b^2=\frac{a^2}{2}+\frac{b^2}{2}\Rightarrow a^2=3b^2\Rightarrow e^2=\frac{3b^2-b^2}{3b^2}=\frac{2}{3}$$

63. $$2\sin 2x\cos x = 2\left(2\cos^2 x\right)$$
$$\Rightarrow 4\cos^2 x(\sin x-1)=0\Rightarrow \sin^2 x=1$$

64. $$I=\int_0^{\pi/4}\frac{(\tan x)\,dx}{\tan x+\tan\left(\frac{\pi}{4}-x\right)}\Rightarrow 2I=\frac{\pi}{4}\Rightarrow I=\frac{\pi}{8}$$

65. $$\tan^{-1}y=y\Rightarrow\frac{dy}{y}+\frac{(dx)(2x)}{1+x^2}=0$$
$$\Rightarrow y\left(1+x^2\right)=C$$
$$\Rightarrow\left(\tan^{-1}y\right)=\frac{1}{1+x^2}$$
$$\Rightarrow y=\tan\left(\frac{1}{1+x^2}\right)$$
$$\lim_{x\to 0}\frac{\tan\left(\frac{1}{1+x^2}\right)-\tan(1)}{x^2}=\sec^2\left(\frac{1}{1+x^2}\right)\times\frac{-2x}{\left(1+x^2\right)^2}$$
$$=-\sec^2(1)$$
$$\Rightarrow y=\tan\left(\frac{1}{1+x^2}\right)$$
$$\tan^{-1}\left(f(1)\right)=\frac{1}{2}$$

66. $$M^9=PQ^9P\Rightarrow \text{tr}\left(M^9\right)=\text{tr}\left(Q^9\right)=3^9+(-3)^9+(2)^9=512$$

67. $$\begin{vmatrix}2&-3&1&4\\1&-3&2&3\\1&3&-p&-q\end{vmatrix}\Rightarrow p=4;q=1$$
$$\Rightarrow\frac{5^{50}}{16}=\frac{(624+1)^{12}(16+9)}{16}\Rightarrow\lambda+\frac{9}{16}$$
$$\Rightarrow \text{Remainder}=9$$

68. $$\text{Mean}=\frac{f_1+2f_2+3f_3}{f_1+f_2+f_3}=2+\frac{f_3-f_1}{f_1+f_2+f_3}\geqslant 2 \text{ (Equality hold when)}$$
$$f_3=f_1=2$$

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69. $(1,1)\ \&\ (3,5)$ is to be added.

70. Given $\vec{a}.\vec{b}=\left|\vec{a}\right|\left|\vec{b}\right|\cos\left(60^0\right)=(2)(3)(1/2)=3$. From $\vec{p}\times\vec{b}=\vec{a}\times\vec{b}$,

we have $\left(\vec{p}-\vec{a}\right)\times\vec{b}=\vec{0}$, so $\vec{p}=\vec{a}+\lambda\vec{b}$.

Using $\vec{p}.\vec{a}=0$, we get $4+3\lambda=0\Rightarrow\lambda=-4/3$. Thus $\vec{p}=\vec{a}-\frac{4}{3}\vec{b}$.

Similarly, $\vec{b}\times\vec{a}\Rightarrow\vec{q}=\vec{b}+\mu\vec{a}$. using $\vec{q}.\vec{b}=0$ we get $9+3\mu=0\Rightarrow\mu=-3$. Thus, $\vec{q}\times\vec{b}-3\vec{a}$.

$$\vec{p}.\vec{q}=\left(\vec{a}-\frac{4}{3}\vec{b}\right).\left(\vec{b}-3\vec{a}\right)=5\left(\vec{a}.\vec{b}\right)-3\left|\vec{a}\right|^2-\frac{4}{3}\left|\vec{a}\right|^2$$

$\vec{p}.\vec{q}=15-12-12=-9$

71. $$3\left[\frac{f(1)+f(-1)}{2}\right]+2\left[\frac{f(1)-f(-1)}{2}\right]$$

$$=\frac{5f(1)}{2}+\frac{f(-1)}{2}=\frac{5}{2}\left(3^{20}\right)+\frac{1}{2}$$

72. $f(0)=0$

$$\Rightarrow f'(x)=k+3x^2-2x$$

$$\Rightarrow f'(x)=3x^2-2x \text{ and } f(x)=x^3-x^2$$

73. $$\lim_{x\to\infty}=\frac{(\sqrt{5}+1)^2(\sqrt{5}+1)^x-(\sqrt{5}-1)^{x+2}}{(\sqrt{5}+1)^x-(\sqrt{5}-1)^x}=6+2\sqrt{5}$$

$$\left[\sqrt{6+2\sqrt{5}}\right]=\left[\sqrt{5}+1\right]=3$$

74. $D\geq 0\Rightarrow p(p-6)\geq 0$

$$f(-6)>0\ \&\ f(1)>0\ \&\ -6<\frac{\alpha+\beta}{2}<1$$

$$\Rightarrow 2<p<9$$

$$\Rightarrow 6\leq p<\frac{27}{4}\Rightarrow p=6$$

$$\Rightarrow g_r g_{21-r}=2\times 6=12$$

75. (i) $x+\frac{5}{2}>1\Rightarrow\left(\frac{x-5}{2x-3}\right)^2>\left(x+\frac{5}{2}\right)^0$

$\Rightarrow 3x^2-2x-16<0$

$x>\frac{-3}{2}$ and $x\in\left(-2,\frac{8}{3}\right)$

$\therefore$ 4 solutions

(ii) $0<x+\frac{5}{2}<1$, No integral solution.

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